

EULER CIRCLE PSET COMPILATION

Problem 1. In how many ways can we place $k \leq \min(m, n)$ identical rooks on an $m \times n$ chessboard in such a way that no two attack each other? (That is, no two are allowed to be in the same row (rank) or column (file).) Your answer should be expressed in terms of k , m , and n .

Solution. WLOG $\min(m, n) = n$. We know that $k \leq n$ and we will look at each case. We place the i th rook in the i th column where $i \leq n$.

Case 1: $k = n$

Note that in every column there must be a rook. The first rook has m options, the second has $m - 1$ options and so on till the n th rook has $m - n + 1$. So totally we have $m(m - 1) \dots (m - n + 1) = \binom{m}{n}$.

Case 2: $k = n - 1$

Note that exactly one column is going to be filled. This can be chosen in n ways. So we are left with an m by $n - 1$ grid and we place the rooks in $\binom{m}{n-1}$ ways by the same logic shown above. Thus the total way is $\binom{n}{1} \binom{m}{n-1}$.

Case 3: $k = n - 2$

Note that exactly two columns are going to be filled. This can be chosen in $\binom{n}{2}$ ways. We now have a m by $n - 2$ grid and we place the rooks in $\binom{m}{n-2}$ ways by similar logic shown above. Thus totally we have $\binom{n}{2} \binom{m}{n-2}$.

$\vdots$

Case i : $k = n - i + 1$

Note that exactly $i - 1$ columns are going to be filled. We now have a m by $n - i + 1$ grid and we place the rooks in $\binom{m}{n-i+1}$ by similar logic as shown above. Thus totally we have $\binom{n}{i-1} \binom{m}{n-i+1}$.

$\vdots$

Case n : $k = 1$

Note that exactly $n - 1$ columns are going to be filled. We now have a m by 1 grid and we place the rooks in $\binom{m}{1}$ by similar logic as shown above. Thus the totally we have $\binom{n}{n-1} \binom{m}{1}$.

So we have $\sum_{k=1}^n \binom{n}{n-k} \binom{m}{k}$. By Vandermonde's we get this to be $\boxed{\binom{m+n}{n}}$.

Problem 2. Consider an equilateral triangle of sidelength n , broken down into equilateral triangles of sidelength 1, as shown in Figure 3 when $n = 5$. How many paths are there from the top triangle to the middle triangle in the bottom row, such that adjacent triangles in the path share a common edge and never go up or revisit a triangle? One such path is shown in Figure 3.

Solution. We note that every path starting from the top and going to the bottom is the same as starting from the bottom and going to the top. Also note that we can only jump to the next row if and only if we are on a triangle with an edge on the second column.

Definition: Call a triangle good if share an edge with the row above it.

Assume we are on a n by n equilateral triangle grid. Denote row k as the row with k equilateral triangles. Row n has $n - 1$ good triangles. We can choose any of these $n - 1$ triangles. Once we do so, we note that there is exactly 1 way to get to that triangle. Now we move up to row $n - 1$. Clearly row $n - 1$ has $n - 2$ good triangles. Again we choose any of these $n - 2$ triangles to move up to the next level. We continue this process again and again till we reach row 1 where our process will terminate.

Thus the total ways is $(n-1)(n-2)(n-3)\dots 1 = \boxed{(n-1)!}$.

Problem 3. Prove that if all the vertices of a bipartite graph have the same nonzero degree, then it has a perfect matching.

Solution. Claim: If G is a bipartite graph with parts A and B the

$$\sum_{v \in A} \deg(v) = \sum_{v \in B} \deg(v).$$

Base Case: $|A| = |B| = 1$. This is clearly as there is only one edge.

Inductive Step: Take a subset of G which has some vertices of G . By drawing one edge from A to B we get that $\sum_{v \in A} \deg(v) = \sum_{v \in B} \deg(v) = 1$. Assume this hold for n vertices. By drawing one more edge we increase the degree on both sides by 1. So the claim holds.

This means that both of our set contain an equal number of elements as all the vertices have the same degree. So this means that there will be $n \cdot |S|$ edges from A to B . Assume $|N(S)| < |S|$. This means that $n \cdot |N(S)|$ edges are leaving from B to A which is a contradiction since $n \cdot |N(S)| < n \cdot |S|$. Therefore $|N(S)| \geq |S|$ and by Hall Marriage Theorem a perfect matching always exists.

Problem 4. We are given two square sheets of paper with integer area n . Suppose we divide each of these papers into n regions, each of area 1.

Solution. Let the papers represent the two parts of a bipartite graph. Let the first sheet of paper represent the part A and the second sheet of paper represent the part B . We connect the two vertices from both parts iff the polygons have a non-empty intersection when placed on top of each other. We are trying to prove that a perfect matching exists. Consider $S \subseteq A$. It's neighbor set has $N(S)$. Note that S is a subset of $N(S)$. Thus $|N(S)| \geq |S|$. So by Hall's Marriage Theorem, a perfect matching exists.

Problem 5. Suppose $G = (V, E)$ is a bipartite graph, where X and Y are the parts of V , and that every vertex has degree ≥ 1 . Suppose that for every edge $e = \{x, y\}$, where $x \in X$ and $y \in Y$, we have $d(x) \geq d(y)$. Show that $|X| \leq |Y|$.

Solution. Since G is a bipartite graph we have

$$\sum_{v \in A} \deg v = \sum_{v \in B} \deg v.$$

Since $\deg(x) \geq \deg(y)$ we know that there must be less elements in X than the number of elements in Y . Thus $|X| \leq |Y|$.

Problem 6. A permutation of a set S is a bijection $f : S \rightarrow S$. A fixed point of a permutation f is some $s \in S$ such that $f(s) = s$. Let $p_k(n)$ be the number of permutations of $[n]$ with exactly k fixed points. Prove that

$$\sum_{k=0}^n k p_k(n) = n!.$$

Solution. Suppose we have a permutation of $(1, 2, \dots, n)$ where $e_1, e_2, \dots, e_k$ are fixed points. Counting this is the same as counting the number of permutation of $[n]$ where e_1 is the fixed point, the number of permutation of $[n]$ where e_2 is a fixed point and so on. This is nothing but $\sum_{k=0}^n kp_k(n)$.

We can count the number of fixed point in a permutation of $[n]$ in another way. We choose an element e from $[n]$ there are $(n-1)!$ ways to arrange the other numbers. There are n ways to choose e . Thus the total number of fixed permutations is $n!$. So

$$\sum_{k=0}^n kp_k(n) = n!.$$

Problem 7. There are n points in the plane such that no three of them are collinear. Prove that the number of triangles whose vertices are chosen from these n points and whose area is 1, is at most $\frac{2}{3}(n^2 - n)$. (Hint: given two points, how many possible third points can there be such that the triangle they form has area 1?)

Solution. Note that for any two points there can be exactly 4 other points where the area of the triangle is 1. This because we can't have 3 collinear points, and we can have 2 points on both sides of the line. So in our graph of n vertices, we can choose $\binom{n}{2}$ line segments and each have 4. But note that each line segments is counted 3 times for each triangle. Thus the maximum is

$$\frac{4}{3} \binom{n}{2} = \frac{2}{3}(n^2 - n).$$

Problem 8. Given nonnegative integers k, n and r , how many solutions are there to the equation

$$a_1 + a_2 + \dots + a_k = n$$

where each a_i is a nonnegative integer greater than or equal to r ?

Solution. Since every integer is greater than r we can write the equation as $(a'_1 + r) + (a'_2 + r) + \dots + (a'_k + r) = n$ for non-negative integers a'_i . This means that $a'_1 + a'_2 + \dots + a'_k = n - rk$ and this can be solved by regular stars and bars. Since there are $k-1$ we know that the number of ways to do this is

$$\binom{n - rk + k - 1}{k - 1} = \binom{n - k(r - 1) - 1}{k - 1}.$$

Problem 9. Prove that if $\gcd(a, b) = 1$, then $\binom{a+b}{a}$ is divisible by $a + b$. Show that this is not necessarily the case if $\gcd(a, b) \neq 1$.

Solution. This is the same as $\frac{\gcd(a, a+b)}{a+b} \cdot \binom{a+b}{a}$. By Bezouts we can write this as $\frac{ax + (a+b)y}{a+b} \cdot \binom{a+b}{a}$. Let $a + b = n$. So we want to prove that $\frac{ax + ny}{n} \cdot \binom{n}{a}$ is an integer. We already see that the $y \cdot \binom{n}{a}$ is an integer so we only worry about the x . But by expanding we get that this is nothing but $x \binom{n-1}{a-1}$. This $\frac{1}{a+b} \cdot \binom{a+b}{a}$ is an integer.

Problem 10. Prove that for all integers $n \geq 2$, we have

$$\frac{2^{2n-1}}{n} < \binom{2n}{n} < 4^n.$$

Solution. We will prove that $2^{2n} < 2n \binom{2n}{n}$. Note that $\binom{2n}{n} = \binom{n}{0}^2 + \binom{n}{1}^2 + \cdots + \binom{n}{n}^2$. By Cauchy-Schwarz we see that

$$(1 + 1 + \cdots + 1) \left(\binom{n}{0}^2 + \binom{n}{1}^2 + \cdots + \binom{n}{n}^2 \right) \geq \left(\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n} \right)^2 = 2^{2n}$$

This gives $2n \binom{2n}{n} > (n+1) \binom{2n}{n} \geq 2^{2n}$ which implies that $\binom{2n}{n} \geq \frac{2^{2n-1}}{n}$. We then need to prove $\binom{2n}{n} < 4^n = 2^{2n}$. Note that 2^{2n} is counting the number of subsets with a set of $2n$ elements. On the other hand, $\binom{2n}{n}$ counts the number of subset with size n . Since this is already included in 2^{2n} we know that $\binom{2n}{n} < 2^{2n}$. So we can string the inequalities together and we get $\frac{2^{2n-1}}{n} < \binom{2n}{n} < 4^n$.

Problem 11. Evaluate the sum

$$\sum_{i=0}^n \binom{n}{i} d(i)$$

in closed form, where $d(i)$ denotes the number of derangements of i .

Solution. Note that $\binom{n}{i} = \binom{n}{n-i}$ counts the number of fixed points in a random permutation of $[n]$. The $d(i)$ counts the number of ways to scramble the rest. So when we are summing from $i = 0$ to $i = n$ we account for all permutations of $[n]$. Thus the answer is $n!$.

Problem 12. Write down and prove a closed formula for $J(n)$, where J is as in the Josephus problem.

Solution. We claim that the answer is $2^x + y$ where $2^x \leq n$ and $y < 2^x$.

Proof. We will prove this by strong induction. The base case is $J(1) = 2(0) + 1 = 1$ which holds. Assume that the condition holds for all numbers under $2^x + y$.

Case 1: y is even

If y is even, we can say that $J(2^x + y) = 2J(2^{x-1} + \frac{y}{2}) - 1 = 2(\frac{2y}{2} + 1) - 1 = 2y + 1$.

Case 2: y is odd

Let $y - 1 = z$. If y is odd, we can say that $J(2^x + y) = 2J(2^{x-1} + \frac{z}{2}) + 1 = 2(\frac{2z}{2} + 1) + 1 = 2z + 3 = 2(z + 1) + 1 = 2y + 1$.

Since we have proved both cases, we have proved our claim. □

Problem 13. Write down a recurrence for the modification of the Josephus function, where every third person has to sit down.

Solution. Let $J(n)$ denote the safe seat with n people. Denote the n people with numbers $1, 2, \dots, n$. Clearly 3 is going to be killed, then 7 and so on. Note that once 3 is killed we have $n - 1$ people, and we start with 4. This is basically $J(n - 1) + 3$ because instead of starting with 1, you are starting with 4. But we need to keep $(\text{mod } n)$ because we don't want a $n + 3$ rd person. So we get the answer as

$$J(n) = J(n - 1) + 3 \pmod{n}.$$

Problem 14. Let f be an arbitrary function defined on the nonnegative integers, and define g by the formula

$$g(n) = \sum_{k=0}^n \left\lfloor \frac{n}{k} \right\rfloor f(k).$$

Show that we can recover f and g via the formula

$$f(n) = \sum_{k=0}^n (-1)^{n-k} \left\{ \frac{n}{k} \right\} g(k).$$

Solution. We will plug the sum form of $g(k)$ into the sum for $f(n)$. So we have

$$\begin{aligned} & \sum_{k=0}^n (-1)^{n-k} \left\{ \frac{n}{k} \right\} \sum_{i=0}^k \left[\frac{k}{i} \right] f(i) \\ & \sum_{k=0}^n \sum_{i=0}^k (-1)^{n-k} \left\{ \frac{n}{k} \right\} \left[\frac{k}{i} \right] f(i) \\ & \qquad \qquad \qquad f(n) \end{aligned}$$

Thus we are done.

Problem 15. Give a combinatorial proof of the identity

$$\left\{ \frac{n+1}{m+1} \right\} = \sum_{k=0}^n \binom{n}{k} \left\{ \frac{k}{m} \right\}.$$

Solution. Suppose we start off with $m+1$ boxes. We will choose k objects out of the n objects and arrange them into m boxes in $\left\{ \frac{k}{m} \right\}$. Then $n-k$ objects can go into the last box. Now, notice that $n-k$ can range from 0 to n . We are just looking for the number of ways to partition $n+1$ objects into $m+1$ boxes. This can be done in $\left\{ \frac{n+1}{m+1} \right\}$. Thus we have proven the result.

Problem 16. Show $\mathbb{R}$ is countable.

Solution. Take two arbitrary digits a and b where $a, b \in [1, 8]$. Now construct a real number r_1 with a random combination of a and b 's. Similar we can construct k numbers in the same fashion. Let $f_{r_i}(k)$ represent the k th digits of the r_i th real number. So if $f_{r_1}(1) = a$, then we are going to construct our new number n such that $f_n(1) = b$. Similarly if $f_{r_i}(i) = a$ we know that $f_n(i) = b$ and if $f_{r_i}(i) = b$ we know that $f_n(i) = a$. Thus we have created a new number n which isn't part of the r_i number set. So $\mathbb{R}$ is countable.

Problem 17. Prove that the closed unit square $[0, 1] \times [0, 1]$ has cardinality $\mathfrak{c}$.

Solution. There exists an injection from $[0, 1] \times [0, 1]$ and R . We want to prove that there exists a injection from R to $[0, 1]$. Consider the function $f(x) = \sin(x)$. Clearly there exists a surjection between $[0, 1] \times [0, 1]$ and R which means there exists a injection from R to $[0, 1] \times [0, 1]$ since there exists a injective function $g : T \rightarrow S$ if and only if $f : S \rightarrow T$ is surjective. So by CSB we are done.

Problem 18. What is the cardinality of the set of functions from the integers to the integers?

Solution. We know that $|\mathbb{Z}| = |\mathbb{N}|$. So the given set has cardinality N^N . Clearly $N^N \geq 2^N$ and $N^N \leq (2^N)^N = 2^{(N^2)} = 2^N$. So $N^N = 2^N$, which means it has cardinality of R .

Problem 19. Prove that for every $x \in \mathbb{Q}$ other than $\langle 0, 1 \rangle$, there is exactly one $y \in \mathbb{Q}$ such that $xy = 1$, where $1 = \langle 1, 1 \rangle$.

Solution. Let $x = \langle a, b \rangle$ and $y = \langle c, d \rangle$. So $xy = \langle a, b \rangle \cdot \langle c, d \rangle = \langle ac, bd \rangle$. We want this to equal to 1, so we know that $ac = bd$ which means that $\langle a, b \rangle = \langle d, c \rangle \Rightarrow x = \langle d, c \rangle$. So the only y that satisfies this is $y = \langle b, a \rangle = \frac{1}{x}$.

Problem 20. Prove that if $x = (A, B)$ is a Dedekind Cut, then $x = \sup(A)$.

Solution. By convention we choose not to include x in A . Let $\alpha \in A$. We know that the $\lim_{\alpha \rightarrow x} \alpha = x$. But since we have decided not to include x in A we know that the value of α will never quite reach x . Thus the $\sup(A) = x$.

Problem 21. Prove the Greatest Lower Bound Property: if $S \subseteq \mathbb{R}$ is bounded below and nonempty, then S has a greatest lower bound.

Solution. We can prove this directly from the Least Upper Bound Property. Consider the a set $T = \{-x | x \in S\}$. Clearly this set T is bounded above. Let $\sup(T) = a$. Then $\inf(S) = -a$ as desired.

Problem 22. Which of the following sequences in $\mathbb{Q}$ are Cauchy sequences? Which ones converge in $\mathbb{Q}$? For those that converge, determine their limits. In all cases, prove that your answers are correct.

(a) $a_n = 1 - \frac{1}{n^3}$ (b) $a_n = \frac{2n^2}{n^2+1}$ (c) $a_n = \frac{n^2}{n+6}$

Solution.

(a)

We claim that this sequence is Cauchy.

Suppose $m, n > N$. Then

$$|a_m - a_n| = \left| \frac{1}{n^3} - \frac{1}{m^3} \right| < \max \left(\frac{1}{n^3}, \frac{1}{m^3} \right) < \frac{1}{N^3},$$

whenever $m, n > N$. Thus it suffices to choose an N such that $\frac{1}{N^3} < \epsilon$ or $N^3 > \frac{1}{\epsilon}$. When that happens we have

$$|a_m - a_n| = \left| \frac{1}{n^3} - \frac{1}{m^3} \right| < \max \left(\frac{1}{n^3}, \frac{1}{m^3} \right) < \frac{1}{N^3} < \epsilon$$

which is just what we wanted.

We claim that this series converges to 1 as n approaches ∞ . In other words we want to prove that $\lim_{n \rightarrow \infty} 1 - \frac{1}{n^3} = 1$.

Let $\epsilon > 0$ and let $N = \sqrt[3]{\frac{1}{\epsilon}}$. Then for all $n > N$,

$$\left| 1 - \frac{1}{n^3} - 1 \right| = \left| \frac{1}{n^3} \right| = \frac{1}{n^3} < \frac{1}{N^3} = \epsilon.$$

Hence, our claim.

(b)

We claim that this series converges to 2 as n approaches ∞ . In other words, we want to prove that

$$\lim_{n \rightarrow \infty} \frac{2n^2}{n^2+1} = 2.$$

Let $\epsilon > 0$ and let $N = \sqrt{\frac{2}{\epsilon}}$. Then for all $n > N$,

$$\left| \frac{2n^2}{n^2+1} - 2 \right| = 2 \left| \frac{n^2}{n^2+1} - 1 \right| = 2 \left| \frac{1}{n^2+1} \right| = \frac{2}{n^2+1} < \frac{2}{N^2} < \epsilon.$$

Hence, our claim.

Since all convergent sequences are also Cauchy (this can be shown by using the triangle inequality and letting $\epsilon' = \frac{\epsilon}{2}$) we know that this sequence is also Cauchy.

(c)

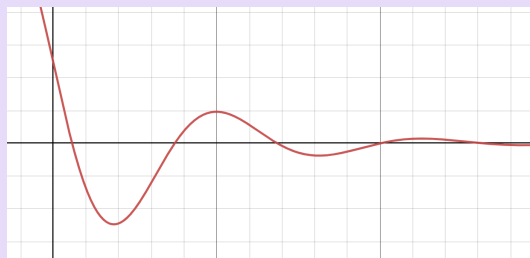
We claim that this sequence is not Cauchy.

Consider $\epsilon = 1$ and $m, n = 5, 1$ respectively. For this sequence to be Cauchy, we know that $|a_m - a_n| < \epsilon$ for all $\epsilon > 0$ i.e. $|a_5 - a_1| = \frac{25}{11} - \frac{1}{7} = \frac{164}{77} \not< 1$.

Since all convergent sequences are Cauchy, and this sequence is not Cauchy, we can conclude that this sequence doesn't converge.

Problem 23. If a_n is a convergent sequence in $\mathbb{Q}$, is $b_n = \lfloor a_n \rfloor$ necessarily convergent? Prove or find a counterexample

Solution. Consider the function $y(t) = Ae^{\frac{-bt}{2m}} \cos(\omega t + \phi)$. This function represents a damped oscillation.



At each point a_n on or above the x axis, we see that $b_n = 0$ and at each point a_n below the x axis we see that $b_n = -1$. Clearly the sequence b_n isn't convergent while a_n converges to 0.

Problem 24. A sequence $a = (a_1, a_2, a_3, \dots)$ is said to be monotonically increasing if $a_1 \leq a_2 \leq a_3 \leq \dots$ and monotonically decreasing if $a_1 \geq a_2 \geq a_3 \geq \dots$. We say that a is monotone if it is either monotonically increasing or monotonically decreasing. Prove that every sequence in $\mathbb{R}$ contains a monotone subsequence.

Solution. Let our sequence be $a = (a_1, a_2, \dots)$. Define a peak at a_n if for every $m \geq n$, $a_n > a_m$. There are two cases to consider: infinite peaks and finite peaks.

Infinite Peaks: Let a_{n_k} be the k th peak. By definition we know that $a_{n_1} \geq a_{n_2}$ since $n_2 \geq n_1$ and we can repeat the same process again. So we get $a_{n_1} \geq a_{n_2} \geq a_{n_3} \geq \dots$ which is a decreasing monotonic sequence.

Finite Peaks: Let the last peak be a_n . Now we see that a_{n+1} can't be a peak. This means that

there is some term $a_{n+k} > a_{n+1}$ since a_{n+1} isn't a peak. Again there there is going to be some term $a_{n+k+j} > a_{n+k}$ since a_{n+k} isn't a peak. Thus we have a sequence $a_n > a_{n+1} > a_{n+k} > \dots$ and this is monotonically increasing.

Problem 25. Prove that a monotone sequence is convergent if and only if it is bounded.

Solution. We will first prove that a monotone sequence is convergent given that it is bounded.

Let $a = (a_1, a_2, \dots)$ be an increasing sequence and bounded above. Let $L = \sup(a)$. For any $\epsilon > 0$, by the definition of supremum, we know that there exists some n such that $a_n < L + \epsilon$. We pick k such that $k \geq n$ and since $a_n < L + \epsilon$ and by the definition of increasing we know that $a_k \geq a_n$ and by definition of supremum we know that $a_k < L$. Now we can string all these to get and we get $L - \epsilon < a_n \leq a_k < L + \epsilon$ and this means that the sequence is convergent. A similar argument can be applied when the sequence is decreasing and bounded below using the infimum.

Now we will prove that a monotone sequence is bounded if it is convergent.

By definition of convergent we know that $|a_n - a| < \epsilon$. Take $\epsilon = 1$ and for all $n > N$ and this gives

$$a - 1 < a_n < 1 + a.$$

So the upperbound is just the $U = \max(a_1, a_2, \dots, a_N, 1 + a)$ and the lower bound is just the $L = \min(a_1, a_2, \dots, a_N, a - 1)$.

Problem 26. Let $\{\mathcal{T}_\alpha : \alpha \in A\}$ be a collection of topologies on X . Prove that $\bigcap_{\alpha \in A} \mathcal{T}_\alpha$ is also a topology on X .

Solution. We will just go through each condition and verify if they are true. For the sake of names, we will call $\mathcal{T} = \bigcap_{\alpha \in A} \mathcal{T}_\alpha$.

- $\emptyset$ and X are in every topology. So they are in $\mathcal{T}$.
- Now we have to check the arbitrary unions of A_i where A_i are the elements of $\mathcal{T}$. Consider $\bigcup_i A_i$. Clearly this is open since each A_i is in the intersection so the union.
- Now we have to check the arbitrary intersections of A_i are in $\mathcal{T}$. Clearly the intersection of arbitrary A_i is open so it is open in $\mathcal{T}$.

Clearly these meet the requirements and hence $\mathcal{T}$ is a topology on X .

Problem 27. Let $f : X \rightarrow Y$ be a function. Then

- If $V \subseteq Y$, then $f(f^{-1}(V)) \subseteq V$, and
- If $U \subseteq X$, then $U \subseteq f^{-1}(f(U))$.

Solution. Take an arbitrary element $v \in f(f^{-1}(V))$. Then by definition

$$v \in \{y \in Y : y = f(x) \text{ for some } x \in f^{-1}(V)\}.$$

So $v = f(x)$ for some $x \in f^{-1}(V)$. Note that $f^{-1}(V)$ is made of up $x \in X$ such that $f(x) = v'$ for some $v' \in V$. Since f is a function $v' = v$ and hence $v \in V$.

Take an arbitrary element $u \in U$ and let $y = f(x) \in f(U)$. By definition

$$f^{-1}(f(U)) = \{x \in X : f(x) \in f(U)\}.$$

Since $f(u) \in f(U)$, we know that $u \in f^{-1}(f(U))$.

Problem 28. Define an equivalence relation on $\mathbb{R}$ by $x \sim y$ iff $x - y = 2\pi n$ for some integer n . We can think of the set of equivalence classes as the interval $[0, 2\pi]$ with the endpoints identified, or glued. Prove $\mathbb{R}/\sim$ is homeomorphic to the unit circle $x^2 + y^2 = 1$ in $\mathbb{R}^2$ under the subspace topology of the standard topology on $\mathbb{R}^2$.

Solution. To prove that $\mathbb{R}/\sim$ is homeomorphic to the unit circle $x^2 + y^2 = 1$ in $\mathbb{R}^2$ under the subspace topology, we need to find a function that is bijective, continuous and its inverse is continuous.

Each equivalence class $[x]$ in $\mathbb{R}/\sim$ corresponds to a point on the unit circle where the angle θ is determined by x . So we can map each point x to $(\cos(x), \sin(x))$ on the unit circle. Clearly this map is bijective.

We will now prove that this map is continuous. Open sets on the unit circle, under the subspace topology, are intersections of the unit circle with open sets in $\mathbb{R}^2$. Let U be an arbitrary open set in the unit circle. The inverse image of U under our map is when $(\cos(x), \sin(x)) \in U$. Since U is open in $\mathbb{R}^2$, there exists an open set $V \in \mathbb{R}^2$ such that $U = V \cap (x^2 + y^2 = 1)$. Note that the preimage of V is just the set of all $x \in \mathbb{R}$ such that $(\cos(x), \sin(x)) \in V \cap (x^2 + y^2 = 1)$. So we know that x is in the preimage of $V \cap (x^2 + y^2 = 1)$ in $\mathbb{R} \rightarrow \mathbb{R}/\sim$.

Since the preimage of an open set under a continuous map is open, we have shown that the inverse image of U under our map is open.

Problem 29. Prove that $(0, 1)$, $(0, 1) \cup (1, 2)$, are not homeomorphic and $(0, 1)$, $[0, 1]$ are not homeomorphic.

Solution. We will take a clever shortcut. Instead of checking if they are homeomorphic directly we will check if certain properties such as connectedness and compactness are preserved under the homeomorphism.

Let's consider $(0, 1)$ and $(0, 1) \cup (1, 2)$. Both of the sets are open intervals but note that $(0, 1) \cup (1, 2)$ has two disconnected intervals where as $(0, 1)$ is connected. Since the property of connectedness is preserved under homeomorphisms, we can conclude that $(0, 1)$ and $(0, 1) \cup (1, 2)$ are not homeomorphic.

Now let's consider $(0, 1)$ and $[0, 1]$. The set $(0, 1)$ is an open interval, while $[0, 1]$ is a closed interval. The closed interval $[0, 1]$ is compact, meaning that every open cover of $[0, 1]$ has a finite subcover. But the open interval $(0, 1)$ is not compact. Since compactness is preserved under homeomorphisms, we can conclude that $(0, 1)$ and $[0, 1]$ are not homeomorphic.

Problem 30. Let Y be a closed subset of X . Prove if A is closed in Y then A is closed in X .

Solution. If $Y - A$ is to be open, we must have $Y - A = Y \cap B$ for an arbitrary open set $B \in X$. So we know that $Y - A \subset B$. Since $(X - Y) \cup B$ is open and $(X - Y) \cup B = X - A$ we know that A has to be closed in X .

Problem 31. Prove that $\mathbb{Q}$ is dense in $\mathbb{R}$.

Solution. For definition of dense is if the closure of a subset in the topology X equals X . We want to show that $\overline{\mathbb{Q}} = \mathbb{R}$. Since $\mathbb{Q} \subseteq \mathbb{R}$ by the definition of closure we know, $\overline{\mathbb{Q}} \subseteq \mathbb{R}$. We need to show that all real numbers are a limit point of $\mathbb{Q}$. Let x be a real number. (Not sure how to proceed without using the fact of density).

Problem 32. Let R be $\mathbb{Z}, \mathbb{Z}[i]$, or $\mathbb{Z}[\omega]$. For $a, b \in R$, we define the *greatest common divisor* of a and b , denoted $\gcd(a, b)$, to be an element g of R such that both a and b are multiples of g , and such that if h is any element of R such that a and b are both multiples of h , then $N(h) \leq N(g)$.

(a) Prove that if g_1 and g_2 are both gcds of a and b , then g_1 and g_2 are associates. (b) Computer the following gcds in $\mathbb{Z}[i]$ or $\mathbb{Z}[\omega]$.

(i) $\gcd(55 - 115i, 123 - 239i)$

(ii) $\gcd(45 - 23\omega, 50 + 69\omega)$

Solution. (a) Let's consider prime factorization to solve this problem. If we let two primes be equivalent if they differ by a factor of a unit, then if WLOG we construct the $\gcd(a, b)$ by iterating through each prime in a and checking if it or it's equivalent divides b , then we get some $\gcd(a, b) = u\pi_1\pi_2 \cdots \pi_n$, and since the set of units are closed under multiplication, we can manipulate the gcds into each other through changes in units, hence they're all associates.

(b) (i) Note that these numbers are pretty small, so Euclidean algorithm is a bit overkill. So instead, we'll use the fact that the norm of the gcd must divide the gcd of the norms of the two values. Since $\gcd(N(55 - 115i), N(123 - 239i)) = 50$, we can test something like $5i \pm 5, 7 \pm 1$, and note that $7 - i$ divides both. Now, we have $\gcd(55 - 115i, 123 - 239i) = (7 - i)\gcd(10 - 15i, 22 - 31i)$ Since $\gcd(N(10 - 15i), N(22 - 31i)) = 5$, we can look at $2 \pm i$, and note that

$$(7 - i)\gcd(10 - 15i, 22 - 31i) = (7 - i)(2 - i)\gcd(15 - 8i, 7 - 4i) = 13 - 9i$$

because the norms of $15 - 8i$ and $7 - 4i$ are relatively prime. Thus, the gcd of $55 - 115i$ and $123 - 239i$ is $13 - 9i$ and its associates.

(ii)

$$\gcd(N(45 - 23\omega), N(50 + 69\omega)) = 1 \implies \gcd(45 - 23\omega, 50 + 69\omega) = 1.$$

Problem 33. Prove that for any prime $p > 3$, the numerator of $\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{p-1}$, in lowest terms, is divisible by p .

Solution. Observe that $\frac{1}{k} + \frac{1}{p-k} = \frac{p}{k(p-k)}$. Also

$$S = (p-1)! \sum_{k=1}^{p-1} \frac{p}{k(p-k)} = p \sum_{k=1}^{p-1} \frac{(p-1)!}{k(p-k)} = pT$$

and from this we know that T is an integer. By Wilson's

$$\frac{(p-1)!}{k(p-k)} \equiv (k^2)^{-1} \pmod{p}.$$

Therefore

$$T \equiv \sum_{k=1}^{p-1} (k^2)^{-1} \equiv \sum_{k=1}^{p-1} k^2 \equiv \frac{(p-1)p(2p-1)}{6} \equiv 0 \pmod{p}.$$

Problem 34. Evaluate the sum $\sum_{a=1}^{p-1} g_a$.

Solution. Note that

$$g_a = \sum_{t=0}^{p-1} \left(\frac{a}{p}\right) \zeta^{at} = \left(\frac{a}{p}\right) g_1.$$

So we know that

$$\sum_{a=1}^{p-1} g_a = g_1 \sum_{a=1}^{p-1} \left(\frac{a}{p}\right) = 0$$

since $\sum_{a=1}^{p-1} \left(\frac{a}{p}\right) = 0$.

Problem 35. Let p be an odd prime. Prove that

$$1^2 \times 2^2 \times 3^2 \cdots \times (p-2)^2 \equiv (-1)^{(p+1)/2} \pmod{p}.$$

Solution. Note that

$$1(p-1)(-1) \times 3(p-3)(-1) \times \cdots \times (p-2)(2)(-1) \equiv (p-1)!(-1)^{(p-1)/2} \equiv (-1)^{(p+1)/2}.$$

Problem 36. Prove: If f and g are equivalent binary quadratic forms, then $\text{Disc}(f) = \text{Disc}(g)$.

Solution. Let $f = ax^2 + bxy + cy^2$ and $g = a'x^2 + b'xy + c'y^2$. Let $g = f' \cdot f$ where $f' = \begin{bmatrix} p & q \\ r & s \end{bmatrix}$ and $\det(f') = 1$. So we get
 $a' = ap^2 + bpq + cq^2$
 $b' = 2apr + b(ps + qr) + 2cqs$
 $c' = ar^2 + brs + cs^2$.

Note that

$$\begin{bmatrix} a' & b'/2 \\ b'/2 & c' \end{bmatrix} = \begin{bmatrix} p & q \\ r & s \end{bmatrix} \begin{bmatrix} a & b/2 \\ b/2 & c \end{bmatrix} \begin{bmatrix} p & r \\ q & s \end{bmatrix}.$$

Taking the determinant we get

$$\text{Disc}(g) = -4 \det \left(\begin{bmatrix} a' & b'/2 \\ b'/2 & c' \end{bmatrix} \right) = -4 \det(f')^2 \det \left(\begin{bmatrix} a & b/2 \\ b/2 & c \end{bmatrix} \right) = \text{Disc}(f).$$

Problem 37. If $p \equiv 1 \pmod{4}$ is prime, then we know that there is some integer n such that $n^2 \equiv -1 \pmod{p}$. Explain how to find one. (Hint: Use Wilson's Theorem.)

Solution. Let $p = 2k + 1$. So we know that $(2k)! \equiv -1 \pmod{p}$ by Wilson's Theorem. Note that

$$(2k)! \equiv 1 \cdot 2 \cdot 3 \cdots k \cdot (-k) \cdot (-k-1) \cdots (-1) \equiv -1 \pmod{p}.$$

this can be written as $(1 \cdot 2 \cdot 3 \cdots k)^2 \equiv -1 \pmod{p}$. Hence $n = (1 \cdot 2 \cdot 3 \cdots k) = \left(\frac{p-1}{2}\right)!$.

Problem 38. Which Gaussian integers can be written as the sum of squares of two Gaussian integers? What about Eisenstein integers?

Solution. Note that all odd integers can be expressed as $(k+0i)^2 + (0+(k-1)i)^2 = k^2 - (k^2 - 2k + 1) = 2k - 1$. Also note that all even integers can be expressed as $(k+(k-1)i)^2 + (k-(k-1)i)^2 = k^2 - (k-1)^2 + k^2 - (k-1)^2 = 2(k^2 - (k^2 - 2k + 1)) = 2(2k - 1)$. Hence we have covered all integers.

Problem 39. Prove that the singular cubic curve $y^2 = x^2(x+1)$ has infinitely many integral points. This doesn't violate Siegel's Theorem, because it isn't an elliptic curve.

Solution. Let $z = y/x$. So we know that $z^2 = x + 1$. Let $t \in \mathbb{Q}$ for z . So we know that $x = t^2 - 1$. We know $y = xz = t(t^2 - 1)$. Let $t = p/q$. Therefore we know that $y = (p/q)^3 - (p/q)$ and $x = (p/q)^2 - 1$. Therefore we know that

$$\left(\frac{p^3 - pq^2}{q^3}\right)^2 = \left(\frac{p^2 - q^2}{q^2}\right)^2 \left(\frac{p^2 - q^2}{q^2} + 1\right).$$

Multiplying both sides by q^6 we get

$$(p^3 - pq)^2 = (p^2 - q^2)^2 p^2.$$

Setting $q = 1$ we get a singular cubic curve of the form $(p^3 - 1)^2 = (p^2 - 1)^2((p^2 - 1) + 1)$. Thus $p^3 - p = y$ and $p^2 - 1 = x$ for some integer p . This generates infinitely many solutions.

Problem 40. The right triangle with sides 3, 4, and 5 has area 6. Using elliptic curves, find three more right triangles with rational sides and area 6.

Solution. We know that the elliptic curve describing a triangle with rational sides and area of 6 is of the form $y^2 = x^3 - 36x$. If a solution is (a, b) then the side lengths are of the form

$$\left|\frac{a^2 - 36}{b}\right|, \left|\frac{12a}{b}\right|, \left|\frac{a^2 + 36}{b}\right|.$$

Since $(-3, 9)$ is a solution, this corresponds to the 3-4-5 triangle.

Let P be $(-3, 9)$. So we know that $2P = (25/4, -35/8)$ is another solution. This corresponds to a $7/10, 120/7, 1201/70$ triangle. We can keep on going like this. The algorithm is

- Find the coordinates of kP for some integer k .
- Use the formula shown above to generate a triangle.
- Repeat.

Problem 41. Prove that

$$\frac{n}{\phi(n)} = \sum_{d|n} \frac{\mu^2(d)}{\phi(d)}.$$

Solution. Let $n = p_1^{e_1} p_2^{e_2} \dots p_k^{e_k}$. We know that

$$\begin{aligned} \sum_{d|n} \frac{\mu^2(d)}{\phi(d)} &= \sum_{e_i=0,1} \frac{1}{\phi(p_1^{e_1} p_2^{e_2} \dots p_k^{e_k})} \\ &= \sum_{e_i=0,1} \frac{1}{\phi(p_1^{e_1}) \phi(p_2^{e_2}) \dots \phi(p_k^{e_k})} \\ &= \left(\phi(p_1^0)^{-1} + \phi(p_1)^{-1} \right) \left(\phi(p_2^0)^{-1} + \phi(p_2)^{-1} \right) \dots \left(\phi(p_k^0)^{-1} + \phi(p_k)^{-1} \right) \\ &= \left(1 + \frac{1}{p_1 - 1} \right) \left(1 + \frac{1}{p_2 - 1} \right) \dots \left(1 + \frac{1}{p_k - 1} \right) \\ &= \frac{p_1 p_2 \dots p_k}{(p_1 - 1)(p_2 - 1) \dots (p_k - 1)}. \end{aligned}$$

We know that

$$\frac{n}{\phi(n)} = \frac{n}{n \prod_{p|n} \left(1 - \frac{1}{p} \right)} = \frac{p_1 p_2 \dots p_k}{(p_1 - 1)(p_2 - 1) \dots (p_k - 1)}.$$

Therefore the statement holds.

Problem 42. Prove that

$$\frac{\zeta(s-1)}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\phi(n)}{n^s}.$$

Solution. Let $n = q_1^{r_1} q_2^{r_2} \dots q_j^{r_j}$. By definition

$$\begin{aligned} \frac{\zeta(s-1)}{\zeta(s)} &= \prod_{p \text{ prime}} \frac{(1 - 1/p^s)}{(1 - 1/p^{s-1})} \\ &= \prod_{p \text{ prime}} (1 - p^{-s}) \left(1 + \frac{1}{p^{s-1}} + \frac{1}{p^{2(s-1)}} + \dots \right) \\ &= \prod_{p \text{ prime}} (1 - p^{-s}) \left(1 + \frac{p}{p^s} + \frac{p^2}{p^{2s}} + \dots \right) \\ &= \prod_{p \text{ prime}} \left(1 + \frac{p}{p^s} + \frac{p^2}{p^{2s}} + \dots - \frac{1}{p^s} - \frac{p}{p^{2s}} - \frac{p^2}{p^{3s}} - \dots \right) \\ &= \prod_{p \text{ prime}} \left(1 + \frac{p-1}{p} \left(\frac{1}{p^{s-1}} + \frac{p}{p^{2s-1}} + \frac{p^2}{p^{3s-1}} + \dots \right) \right) \\ &= \prod_{p \text{ prime}} \left(1 + \left(1 - \frac{1}{p} \right) \left(\frac{p}{p^s} + \frac{p^2}{p^{2s}} + \frac{p^3}{p^{3s}} + \dots \right) \right) = \sum \frac{a_n}{n^s}. \end{aligned}$$

So we get that $a_n = \prod_{j=1}^k \left(1 - \frac{1}{q_i} \right) q_j^{r_j}$. This is nothing by $\phi(n)$.

Problem 43. Prove that $\zeta(2) = \frac{\pi^2}{6}$ using double integral method.

Solution. We know that $\int_0^1 \int_0^1 x^{n-1} y^{n-1} dx dy = \frac{1}{n^2}$. Since $\zeta(2) = \sum_{n=1}^{\infty} \frac{1}{n^2}$, we know that

$$\zeta(2) = \sum_{n=1}^{\infty} \frac{1}{n^2} = \sum_{n=1}^{\infty} \int_0^1 \int_0^1 x^{n-1} y^{n-1} dx dy = \int_0^1 \int_0^1 \frac{1}{1-xy} dx dy$$

by the Geometric-Series Formula. Let $u = \frac{x+y}{2}$ and $v = \frac{x-y}{2}$. Then we know that $x = u + v$ and $y = u - v$. Now we can do change of variables to our integral. The determinant of the Jacobian is $\begin{vmatrix} 1 & -1 \\ 1 & 1 \end{vmatrix} = 2$. Note that our old limits of integration are now changed under this transformation hence note that there is a symmetry on the u -axis. Therefore

$$\begin{aligned}\zeta(2) &= 2 \left(\int_0^{1/2} \int_0^u \frac{2dvdu}{1-u^2+v^2} + \int_{1/2}^1 \int_0^{1-u} \frac{2dvdu}{1-u^2+v^2} \right) \\ &= 4 \left(\int_0^{1/2} \frac{1}{\sqrt{1-u^2}} \arctan \left(\frac{u}{\sqrt{1-u^2}} \right) du \right) \\ &\quad + \left(\int_{1/2}^1 \frac{1}{\sqrt{1-u^2}} \arctan \left(\frac{1-u}{\sqrt{1-u^2}} \right) du \right).\end{aligned}$$

Note that $\arcsin(u) = \arctan \left(\frac{u}{\sqrt{1-u^2}} \right)$. Also note that if $\arctan \left(\frac{1-u}{\sqrt{1-u^2}} \right) = x$ then we have that $\tan^2(x) = \frac{1-u}{1+u}$ which implies, by using $\tan^2(x)+1 = \sec^2(x)$, $u = \cos(2x)$. This means that $x = \frac{\arccos(u)}{2}$. To make everything in terms of arcsin we can convert $x = \frac{\arccos(u)}{2} = \frac{\pi}{4} - \frac{\arcsin(u)}{2}$. Substituting this back we get

$$\begin{aligned}\zeta(2) &= 4 \left(\int_0^{1/2} \frac{1}{\sqrt{1-u^2}} \arcsin(u) du \right) \\ &\quad + \left(\int_{1/2}^1 \frac{1}{\sqrt{1-u^2}} \left(\frac{\pi}{4} - \frac{\arcsin(u)}{2} \right) du \right).\end{aligned}$$

Since the anti-derivative of $\frac{1}{\sqrt{1-u^2}} \arcsin(u)$ is $\frac{\arcsin^2(u)}{2}$ we can evaluate the integral to get $\frac{\pi^2}{6}$.

Problem 44. Show that $\log([x]!) = x \log x - x + O(\log x)$.

Solution. Note that $\log(n!) = \log n + \log n - 1 + \dots + \log 1$. Now we can use Euler Summation. Let $f(t) = \log(t)$. Then we have

$$\begin{aligned}\sum \log k &= \int_1^x \log(t) dt + \int_1^x \{t\} \frac{1}{t} dt + \{x\} \log x \\ &= x \log(x) - x + \int_1^x \{t\} \frac{1}{t} dt + O(\log x).\end{aligned}$$

Since

$$0 \leq \int_1^x \frac{\{t\}}{t} dt < \int_1^x \frac{1}{t} dt = \log x$$

is the average order of $O(\log x)$ we know that $\log([x]!) = x \log x - x + O(\log x)$.

Problem 45. Prove that for all integers $n \geq 2$, we have

$$\frac{\sigma(n)}{n} < \frac{n}{\phi(n)} < \zeta(2) \frac{\sigma(n)}{n}.$$

Solution. We will prove that $\sigma(n)\phi(n) < n^2$. By definition

$$\sigma(n) = \prod_{i=1}^k \frac{p_i^{e_i+1} - 1}{p_i - 1} \text{ and } \phi(n) = n \prod_{i=1}^k \left(1 - \frac{1}{p_i}\right).$$

Then we have

$$\sigma(n)\phi(n) = n^2 \prod_{i=1}^k \left(1 - \frac{1}{p_i^{e_i+1}}\right)$$

and hence we have shown the left hand side of the inequality. For the RHS it is not too hard to see that

$$\prod_{i=1}^k \left(1 - \frac{1}{p_i^{e_i+1}}\right)^{-1} < \sum_{n=1}^{\infty} \frac{1}{n^2} < \frac{\pi^2}{6}$$

and hence we have proved the inequality.

Problem 46. Prove the Gamma Duplication Formula of Legendre:

$$\Gamma(2s) = \frac{2^{2s-1}}{\sqrt{\pi}} \Gamma(s) \Gamma(s + 1/2).$$

Solution. We will use the beta function to prove this problem. The Beta function is defined to be

$$\beta(s, w) = \int_0^1 t^{s-1} (1-t)^{w-1} dt.$$

One important thing to note is that $\beta(s, w) = \beta(w, s)$ and this is just through a simple change of variables where t is changed to $1-t$. We will first prove that $\beta(s, w) = \frac{\Gamma(s)\Gamma(w)}{\Gamma(s+w)}$ by using the integral definition of Γ . Note that

$$\Gamma(s)\Gamma(w) = \int_0^{\infty} u^{s-1} e^{-u} du \cdot \int_0^{\infty} v^{w-1} e^{-v} dv.$$

By letting $u = x^2$ and $v = y^2$ we get

$$\Gamma(s)\Gamma(w) = 4 \int_0^{\infty} x^{2s-1} e^{-x^2} dx \cdot \int_0^{\infty} y^{2w-1} e^{-y^2} dy = 4 \int \int_{[0, \infty)^2} x^{2s-1} y^{2w-1} e^{-(x^2+y^2)} dA.$$

Once again we can use change of variables. Let $x = r \sin(\theta)$ and $y = r \cos(\theta)$. This means that $dA = r dr d\theta$ and our bounds change to $r \in [0, \infty)$ and $\theta \in [0, \pi/2]$. So now we have

$$\begin{aligned} \Gamma(s)\Gamma(w) &= 4 \int_0^{\pi/2} \int_0^{\infty} r^{2(s+w)-1} \cos^{2s-1}(\theta) \sin^{2w-1}(\theta) e^{-r^2} r dr d\theta \\ &= 4 \int_0^{\infty} r^{2(s+w-1)} e^{-r^2} r dr \cdot \int_0^{\pi/2} (\cos^2(\theta))^{s-1} (\sin^2(\theta))^{w-1} \cos(\theta) \sin(\theta) d\theta \\ &= 2 \int_0^{\infty} r^{2(s+w-1)} e^{-r^2} r dr \cdot \int_0^{\pi/2} (\cos^2(\theta))^{s-1} (\sin^2(\theta))^{w-1} (2 \cos(\theta) \sin(\theta)) d\theta. \end{aligned}$$

Now we can use change of variables again. Let $m = r^2$. So our integral becomes

$$\Gamma(s)\Gamma(w) = \int_0^\infty m^{(s+w-1)} e^{-m} dm \cdot \int_0^{\pi/2} (\cos^2(\theta))^{s-1} (\sin^2(\theta))^{w-1} (2 \cos(\theta) \sin(\theta)) d\theta.$$

With one last change of variables we let $t = \sin^2(\theta)$. Hence our integral becomes

$$\begin{aligned} \Gamma(s)\Gamma(w) &= \int_0^\infty m^{(s+w-1)} e^{-m} dm \cdot \int_0^1 t^{w-1} (1-t)^{s-1} dt \\ &= \Gamma(s+w)\beta(s, w) \end{aligned}$$

which proves our lemma. Now note that

$$\frac{\Gamma(s)\Gamma(s)}{\Gamma(2s)} = \beta(s, s) = \int_0^1 t^{s-1} (1-t)^{s-1} dt.$$

We can do change of variables again. Let $t = \frac{1+x}{2}$. This means

$$\begin{aligned} \beta(s, s) &= \frac{1}{2} \int -1^1 \left(\frac{1+x}{2} \right)^{s-1} \left(\frac{1-x}{2} \right)^{s-1} dx \\ &= 2^{1-2s} \int_{-1}^1 (1-x^2)^{s-1} dx = 2 \cdot 2^{1-2s} \int_0^1 (1-x^2)^{s-1} dx. \end{aligned}$$

Once again with another change of variables we can let $x = \sqrt{u}$. So our integral becomes

$$\beta(s, s) = 2^{1-2s} \int_0^1 u^{1/2-1} u^{s-1} du = 2^{1-2s} \beta(1/2, s) = 2^{1-2s} \frac{\Gamma(1/2)\Gamma(s)}{\Gamma(s+1/2)}.$$

Since $\Gamma(1/2) = \sqrt{\pi}$, we have

$$\beta(s, s) = \frac{\Gamma(s)\Gamma(s)}{\Gamma(2s)} = 2^{1-2s} \frac{\Gamma(1/2)\Gamma(s)}{\Gamma(s+1/2)}$$

which is what we wanted to prove.

Problem 47. Find

$$\int_0^1 x^{1/3} (1-x)^{2/3} dx.$$

Solution. We will use the β function. By definition we know that $\beta(4/3, 5/3) = \int_0^1 x^{1/3} (1-x)^{2/3} dx$. From our lemma from problem 46, we also know that $\beta(4/3, 5/3) = \frac{\Gamma(4/3)\Gamma(5/3)}{\Gamma(3)}$. We also know that $\Gamma(4/3) = \frac{1}{3}\Gamma(1/3)$ and $\Gamma(5/3) = \frac{2}{3}\Gamma(2/3)$. Hence

$$\begin{aligned} \beta(4/3, 5/3) &= \frac{\Gamma(4/3)\Gamma(5/3)}{\Gamma(3)} \\ &= \frac{2}{9} \cdot \frac{\Gamma(1/3)\Gamma(2/3)}{2} \\ &= \frac{1}{9} \cdot \frac{\pi}{\sin(\pi/3)} \\ &= \frac{2\pi}{9\sqrt{3}} \end{aligned}$$

where the second to last step comes from Euler's Reflection Formula.